

Assignment Kit for Coding Standard



Personal Software Process for Engineers: Part I

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Personal Software Process for Engineers: Part I

Assignment Kit for the Coding Standard

Overview

Overview This assignment kit covers the following topics.

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Prerequisites

Prerequisites

- Read Chapter 4
- Complete Size Counting Standard

Objectives

The objectives of the coding standard are to

- establish a consistent set of coding practices
- provide criteria for judging the quality of the code that you produce
- facilitate size counting by ensuring your programs are written so they can be readily counted
- for LOC counting, require that there be a separate physical line for each logical line of code

Coding standard requirements

Coding standard requirements

Produce, document, and submit a completed coding standard that calls for quality coding practices.

For LOC counting, ensure that a separate physical source line is used for each logical line of code.

Submit the coding standard with your program 2 assignment package.

Example coding Standard

Coding standard example

Pages 5 and 6 of this workbook contain an example C++ coding standard.

Notes about the example

- Since it is an example, tailor it to meet your personal needs.
 - If you have an existing organizational standard, consider using it for the PSP exercises.
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Example C++ Coding Standard

Purpose	To guide implementation of C++ programs
Program Headers	Begin all programs with a descriptive header.
Header Format	<pre> / *****/ /* Program Assignment: the program number */ /* Name: your name */ /* Date: the date you started developing the program */ /* Description: a short description of the program and what it does */ / *****/ </pre>
Listing Contents	Provide a summary of the listing contents
Contents Example	<pre> / *****/ /* Listing Contents: */ /* Reuse instructions */ /* Modification instructions */ /* Compilation instructions */ /* Includes */ /* Class declarations: */ /* CData */ /* ASet */ /* Source code in c:/classes/CData.cpp: */ /* CData */ /* CData() */ /* Empty() */ / *****/ </pre>

(continued)

Example C++ Coding Standard (continued)

Reuse Instructions	- Describe how the program is used: declaration format, parameter values, types, and formats. - Provide warnings of illegal values, overflow conditions, or other conditions that could potentially result in improper operation.
Reuse Instruction Example	<pre> / ***** */ /* Reuse instructions */ /* int PrintLine(char *line_of_character) */ /* Purpose: to print string, 'line_of_character', on one print line */ /* Limitations: the line length must not exceed LINE_LENGTH */ /* Return 0 if printer not ready to print, else 1 */ / ***** */ </pre>
Identifiers	Use descriptive names for all variable, function names, constants, and other identifiers. Avoid abbreviations or single-letter variables.
Identifier Example	<pre> Int number_of_students; /* This is GOOD */ Float: x4, j, ftave; /* This is BAD */ </pre>
Comments	- Document the code so the reader can understand its operation. - Comments should explain both the purpose and behavior of the code. - Comment variable declarations to indicate their purpose.
Good Comment	<pre> If(record_count > limit) /* have all records been processed? */ </pre>
Bad Comment	<pre> If(record_count > limit) /* check if record count exceeds limit */ </pre>
Major Sections	Precede major program sections by a block comment that describes the processing done in the next section.
Example	<pre> / ***** */ /* The program section examines the contents of the array 'grades' and calcu- */ /* lates the average class grade. */ / ***** */ </pre>
Blank Spaces	- Write programs with sufficient spacing so they do not appear crowded. - Separate every program construct with at least one space.
Indenting	- Indent each brace level from the preceding level. - Open and close braces should be on lines by themselves and aligned.
Indenting Example	<pre> while (miss_distance > threshold) { success_code = move_robot (target_location); if (success_code == MOVE_FAILED) { printf("The robot move has failed.\n"); } } </pre>
Capitalization	- Capitalize all defines. - Lowercase all other identifiers and reserved words. - To make them readable, user messages may use mixed case.
Capitalization	<pre> #define DEFAULT-NUMBER-OF-STUDENTS 15 </pre>

Examples	<code>int class-size = DEFAULT-NUMBER-OF-STUDENTS;</code>
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Evaluation criteria and suggestions

**Evaluation
criteria**

Your standard must be

- complete
 - legible
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Suggestions

Keep your standards simple and short.

Do not hesitate to copy or build on the PSP materials.

Coding Standard Template

Purpose	El propósito de este estándar es: • Establecer un conjunto consistente de prácticas de codificación para mis programas Java. • Proveer criterios para evaluar la calidad del código que produzco. • Facilitar el conteo de líneas de código (LOC) siguiendo el estándar PSP.
Program Headers	Begin all programs with a descriptive
Header Format	<pre> /* * <NombreDelArchivo>.java * Autor: Marely Jácome * Fecha: 2025-11-27 * Versión: 1.1 * Descripción: Descripción breve del propósito del archivo. */ </pre>
Listing Contents	Provide a summary of the listing contents.
Contents Example	<pre> /* * Listing Contents: * - Reuse instructions * - Modification instructions * - Compilation instructions * - Dependencies * - Clases y métodos incluidos en el archivo */ </pre>

Reuse Instructions	- Describe how the program is used. Provide the declaration format, parameter values and types, and parameter limits. - Provide warnings of illegal values, overflow conditions, or other conditions that could potentially result in improper operation.
Reuse Example	<pre> /* * Reuse Instructions: * - Esta clase implementa los cálculos de regresión lineal del PSP. * - Requiere arreglos numéricos representados como Strings. * - No acepta valores vacíos o no numéricos. * - Usar setXk(double xk) antes de obtener yk. */ </pre>
Identifiers	Use descriptive names for all variables, function names, constants, and other identifiers. Avoid abbreviations or single letter variables.
Identifier Example	Ejemplos BUENOS (siguiendo el estándar): - EstimacionCorLineal - executeRegression - sumXY - dblSumXX

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Coding Standard Template (continued)

Comments	• Document the code so that the reader can understand its operation. • Comments should explain both the purpose and behavior of the code. • Comment variable declarations to indicate their purpose.
Good Comment	<pre>/** Suma de los valores X usados para promedios y coeficientes */ private double dblSumX; /** Número total de elementos válidos en las listas X y Y */ private int intN; /** Pendiente β_1 calculada en la ecuación de regresión */ private double dblB1;</pre>
Bad Comment	<pre>// convierte texto a arreglo arrDataX = data.saveData(dataX);</pre>
Major Sections	Precede major program sections by a block comment that describes the processing that is done in the next section
Example	SUMATORIAS PSP • PROMEDIOS • PARÁMETROS DE REGRESIÓN • CORRELACIÓN Y r^2 • PREDICCIÓN Yk
Blank Spaces	• Write programs with sufficient spacing so they do not appear crowded. • Separate every program construct with at least one space.
Indenting	• Indent every level of brace from the previous one. • Open and closing braces should be on lines by themselves and aligned with each other.
Indenting Example	<pre>if (x > 0) { y = 2; }</pre>

Capitalization	• Capitalized all defines. • Lowercase all other identifiers and reserved words. • Messages being output to the user can be mixed-case so as to make a clean user presentation.
Capitalization Example	Ejemplo que define en MAYUSCUALAS: <pre>// Correcto: constante totalmente en mayúsculas public static final double DEFAULT_XK = 386.0;</pre> Ejemplo de variable o metodo en minusculas: <pre>// Correcto: identificadores en lowerCamelCase double dblSumX; int intN; public double getAvgX(String[] datalist)</pre> Ejemplo de mensaje para el usuario: <pre>System.out.println("RESULTADOS REGRESIÓN LINEAL PSP"); System.out.println("β0 = " + b0); System.out.println("xk = " + thisXk);</pre>